Data Collection

This Chapter Overview

Interestingly, most of this chapter is from the old S3 module (a Further Maths module!) with also some S2. There is little 'calculation' involved in this chapter; consider this a 'bookwork' one!

1: Populations vs samples

"Suggest why we would not test all the light bulbs."
"Identify the sampling frame."

2: Random Sampling

Describe the disadvantages of systematic sampling.

3: Non-Random Sampling

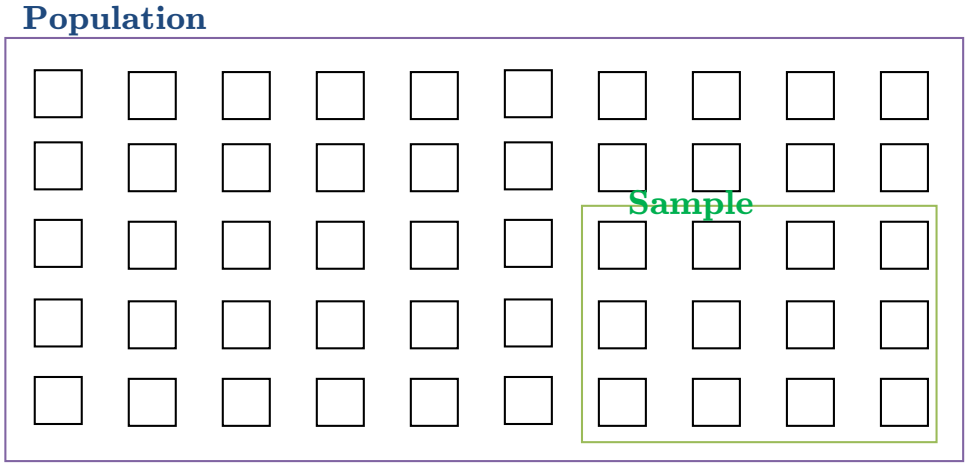
Describe how a stratified sample would be conducted, including strata sizes.

4: Types of data

Continuous vs discrete, terms such as class intervals, class boundaries, class width.

5: Edexcel's 'Large Data Set'

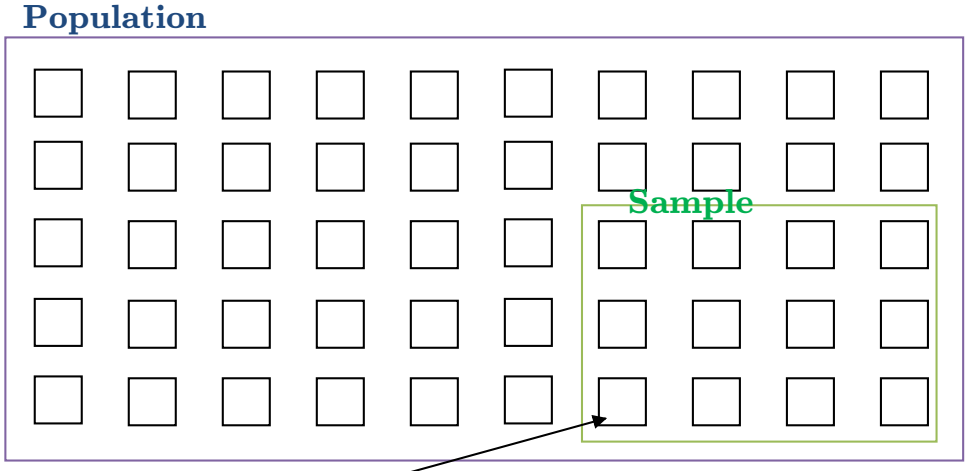
What you're expected to know about the 'large data set' of weather data, and how to use it.



A population is

A sample is

You're probably used to a 'population' meaning all humans/animals within a country/ecosystem. But a population could be "*all the lightbulbs in a factory*" or "*all the cars in the UK*".



Each individual thing in the population that can be sampled is known as a **sampling unit**.

Often sampling units of a population are individually named or numbered **to form a list** called the **sampling frame**.

Populations vs Samples 5

We could collect data either from a sample, or from the entire population.
Data collected from the entire population is known as a

	Advantages	Disadvantages
Census		
Sample		

Example: A supermarket wants to test a delivery of avocados for ripeness by cutting them in half.

a. Suggest a reason why the supermarket should not test all the avocados in the delivery.

b. The supermarket tests a sample of 5 avocados and finds that 4 of them are ripe. They estimate that 80% of the avocados in the deliver are ripe. Suggest one way that the supermarket could improve their estimate.

a a

b b

Exercise 1A – Pearson Statistics Y1 page 3 6

1. A school uses a census to investigate the dietary requirements of its students.
(a) Explain what is meant by a census.
(b) Give one advantage and one disadvantage to the school of using a census.

a

b

2. A factory makes safety harnesses for climbers and has an order to supply 3000 harnesses. The buyer wishes to know that the load at which the harness breaks exceeds a certain figure.

(a) Suggest a reason why a census would not be used for this purpose.

The factory tests four harnesses and records their breaking loads.

320 kg, 260 kg, 240 kg, 180 kg

(b) The factory claims that the harnesses are safe for loads up to 250 kg. Use the sample data to comment on this claim.

(c) Suggest one way in which the company can improve its prediction.

a

b

c

3. A city council wants to know what people think about its recycling centre. It decides to carry out a sample survey of residents.

(a) Write down one reason why the council should not take a census.

(b) Suggest a suitable sampling frame.

(c) Identify the sampling units.

3a

3b

3c

5. A garage manager wants to know what their mechanics think about a new pension scheme designed for them. The manager asks every mechanic in the garage.

(a) Describe the population the manager will use.

(b) Write down the main advantage of asking all the mechanics.

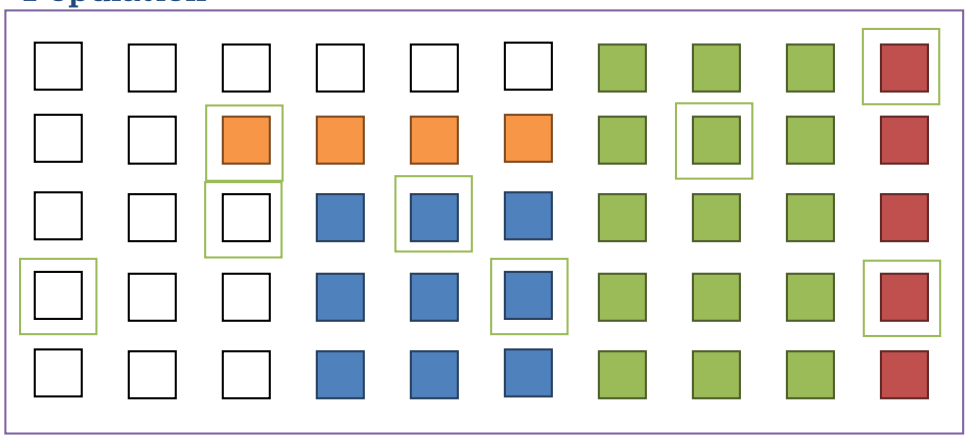
5a

5b

Lay out your notes like this for next bit of the chapter. Use a full page.

	Type	How to carry out	Advantages	Disadvantages
Random Sampling	Simple Random Sampling			
	Systematic Sampling			
	Stratified Sampling			
Non-Random	Quota Sampling			
	Opportunity Sampling			

Population



Ordinarily, we would want each thing in our sampling frame to have an **equal chance of being chosen**, in order to **avoid bias**.

This is known as **random sampling**.
There are a few ways of doing this. . .

Simple Random Sampling 11

Type	How to carry out	Advantages	Disadvantages
Simple Random Sampling			

Edexcel S3 June 2004 Q1a

There are 64 girls and 56 boys in a school. Explain briefly how you could take a random sample of 15 pupils using a simple random sample. **(3)**

Systematic Sampling 12

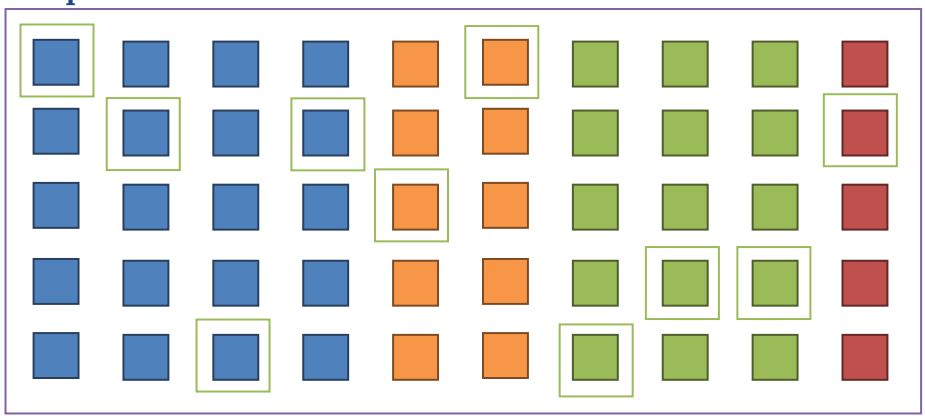
Type	How to carry out	Advantages	Disadvantages
Systematic Sampling			

Edexcel S3 June 2009 Q1a

A telephone directory contains 50 000 names. A researcher wishes to select a systematic sample of 100 names from the directory. Explain in detail how the researcher should obtain such a sample. **(2)**

We want to sample 20% of the population. If the population were divided into distinct groups (e.g. age ranges), known as '**strata**', we could randomly sample 20% from each group, ensuring each group is equally represented.

Population



Type	How to carry out	Advantages	Disadvantages
Stratified Sampling			

Edexcel S3 Jan 2006 Q1

A school has 15 classes and a sixth form. In each class there are 30 students. In the sixth form there are 150 students. There are equal numbers of boys and girls in each class. There are equal numbers of boys and girls in the sixth form. The head teacher wishes to obtain the opinions of the students about school uniforms. Explain how the head teacher would take a stratified sample of size 40. **(7)**

You would certainly want to know your mark scheme on this one!

2. A survey is carried out on 100 members of the adult population of a city suburb. The suburb has 2000 residents. An alphabetical list of the inhabitants is available.

(a) Explain one limitation of using a systematic sample.

(b) Describe a sampling method that would be free of bias for this survey.

2a

2b

3. A gym wants to take a sample of its members. Each member has a five-digit membership number. The gym selects every member whose number ends in 000.

(a) Is this a systematic sample? Give a reason.

(b) Suggest one way to improve the reliability of this sample.

3a

3b

4. A head of sixth form wants the opinions of Year 12 and Year 13 students about the common room facilities. The table shows the numbers in each year.

	Year 12	Year 13
Male	70	50
Female	85	75

(a) Suggest a suitable sampling method for a sample of 40 students.

(b) How many students from each gender in each year should be asked?

4a

4b

5. A factory manager wants to find out how workers travel to work. Each of the 480 workers has a clocking-in number from 1 to 480. Explain how to take a systematic sample of size 30.

5

Exercise 1B – Pearson Statistics Y1 page 6-7 17

6. A sports club director wants a sample of members. Each member has a unique membership number. There are 121 cricket players, 145 hockey players and 104 squash players. No one plays more than one sport.

(a) Explain how to take a simple random sample of 30 and give one disadvantage.

The director decides to take a stratified sample of 30.

(b) State one advantage of this method.

(c) Work out the number of members to select from each sport.

a

b

c

Non-Random Sampling 18

Consider the following scenario: You wish to conduct a survey in the UK **on whether being left-handed affects IQ**. We need to choose people to assess.

Why would random sampling be problematic?

Famous Lefties



For this scenario we'd likely use **quota sampling**, i.e.

1. As with stratified sampling, divide population into groups according to characteristic of interest, then determine size of each group in sample to reflect proportions within the population.
2. But instead of random sampling within each group, we actively choose people within each group via suitable means (e.g. advertising), **until the 'quota' for each group is filled**.

A variant of this is **opportunity sampling**, where we find people **at the same time the survey is being carried out** (e.g. exit polls at polling stations). This is not a suitable method for the left-handed example, because giving the likely time-consuming nature of assessment coupled with resources required, we'd likely arrange with the people taking part before the actual assessment tasks took place.

Quota & Opportunity Sampling 19

Type	How to carry out	Advantages	Disadvantages
Quota Sampling			
Opportunity/ Convenience Sampling			

Example Question 20

Edexcel S3 June 2010 Q2

A lake contains three species of fish. There are estimated to be 1400 trout, 600 bass and 450 pike. A survey of fish health is carried out and a sample of 30 fish is chosen.

(a) Give a reason why stratified random sampling cannot be used. (1)

(b) State an appropriate sampling method for the survey. (1)

(c) Give one advantage and one disadvantage of this method. (2)

(d) Explain how to use this method to select the 30 fish. Show your working. (4)

a

b

c

d

1. Interviewers collect information about spending habits from 40 shoppers in a shopping centre.
(a) Explain how to use (i) quota sampling and (ii) opportunity sampling.
(b) Which method is likely to give a more representative sample?

1a(i)

1a(ii)

1b

2. Describe the similarities and differences between quota sampling and stratified random sampling.

2

3. An interviewer asks the first 50 people he sees outside a fish and chip shop on a Friday evening about their eating habits.
(a) What sampling method did he use?
(b) Why might the sample be unrepresentative?
(c) Suggest two improvements to data collection.

3a

3b

3c

4. A researcher asks the first five people entering a supermarket on Monday morning about their weekly radio listening. They report 4, 7, 6, 8 and 2 hours.
(a) Predict the average weekly hours for the town.
(b) Describe the sampling method and comment on reliability.
(c) Suggest two improvements.

4a

4b

4c

5. Scientists studying the masses of wild deer catch the first five male and the first five female deer they find.

(a) What sampling method are they using?

(b) Give one advantage.

Male (kg)	75	80	90	85	82
Female (kg)	67	72	75	68	65

(c) Compare the masses of male and female deer.

(d) Suggest two improvements to the sampling method.

a

b

c

d

6. The heights in metres of 20 ostriches are listed below.

1.8, 1.9, 2.3, 1.7, 2.1, 2.0, 2.5, 2.7, 2.5, 2.6, 2.3, 2.2, 2.4, 2.3, 2.2, 2.5, 1.9, 2.0, 2.2, 2.5.

(a) Take an opportunity sample of size five.

(b) Starting at the second value, take a systematic sample of five.

(c) Calculate each sample mean.

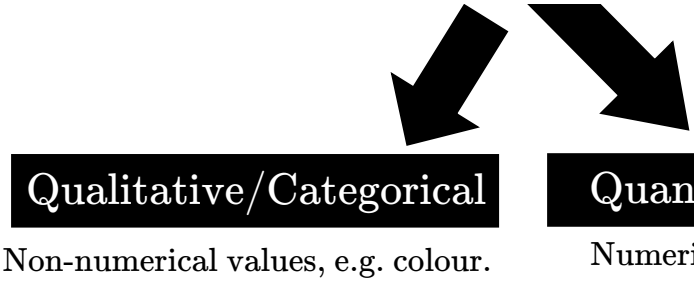
(d) State, with reasons, which method is likely to be more reliable.

a

b

c

d



Qualitative/Categorical

Non-numerical values, e.g. colour.

Quantitative

Numerical values.



Discrete

Can only take specific values, e.g. shoe size, number of children.

Continuous

Can take any decimal value (possible with a specified range).

Note that while discrete variables only allow specific values, the range could still be infinite, e.g. "number of attempts before success".

Weight **w** (kg)	Frequency
$0 \leq w < 20$	3
$20 \leq w < 70$	4

Data can be **grouped** for conciseness, at the expense of losing the exact original values.

This is known as a _____

$20 \leq w < 70$

_____ Midpoint = _____ _____

Class width = _____

Exercise 1D – Pearson Statistics Y1 page 10 26

1. Is each variable qualitative or quantitative?
(a) Height of a tree
(b) Colour of a car
(c) Time waiting in a queue
(d) Shoe size
(e) Names of pupils in a class

1a

1b

1c

1d

1e

2. Is each quantitative variable continuous or discrete?
(a) Shoe size
(b) Length of a leaf
(c) Number of people on a bus
(d) Weight of sugar
(e) Time required to run 100 m
(f) Lifetime in hours of torch batteries

2a

2b

2c

2d

2e

2f

3. Explain why each statement is true.
(a) Type of tree is qualitative.
(b) Number of pupils in a class is discrete and quantitative.
(c) Weight of a collie dog is continuous and quantitative.

3a

3b

3c

4. The table shows the masses of two-month-old lambs.

Mass m (kg)	Frequency
$1.2 \leq m < 1.3$	8
$1.3 \leq m < 1.4$	28
$1.4 \leq m < 1.5$	32
$1.5 \leq m < 1.6$	22

(a) Give the boundaries of the third group.
(b) Find the midpoint of the second group.
(c) Find the width of the first group.

4a

4b

4c

Name That Sampling Method! 28

Simple Random Sampling	Systematic Sampling	Stratified Sampling	Quota Sampling	Opportunity Sampling

Suggest a suitable sampling method.

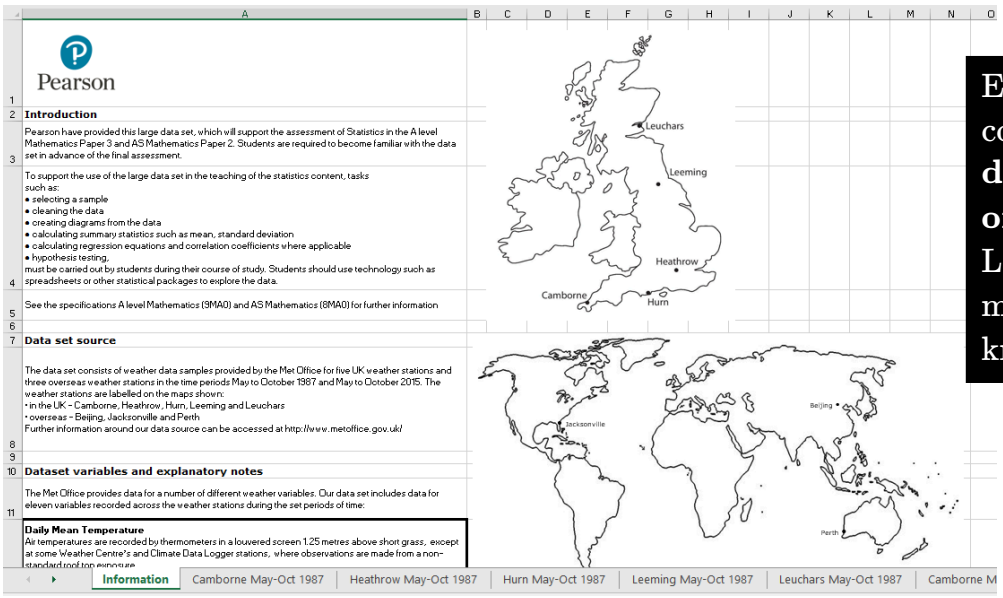
"You wish to test lightbulbs produced by a factory in a daily batch."

"You wish to survey consumer opinion on your new drink *FizzGuzz* released in the UK."

"You wish to determine students' favourite TV programmes in your school, that is fairly representative of each year group."

All A Level exam boards are obligated to provide a 'large data set'. Data in exam questions will often be from this set, and you are encouraged to explore this data (which is publicly available) in Microsoft Excel.

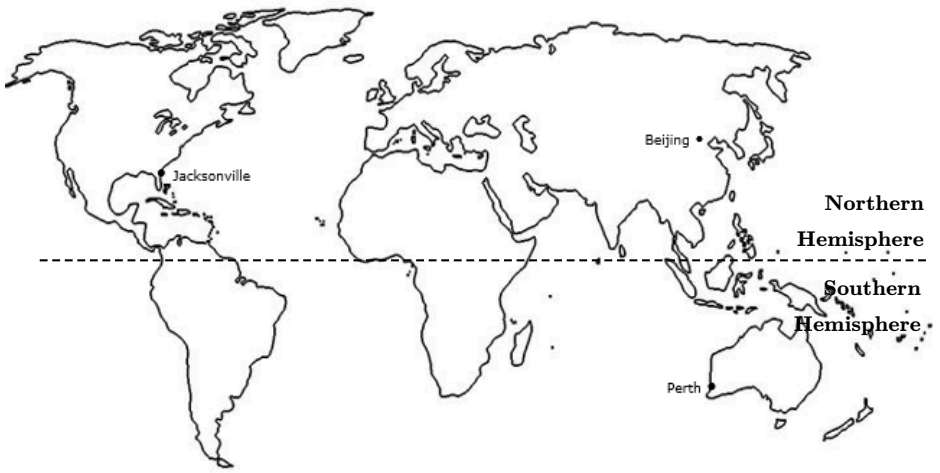
It is important to note that you are expected to be familiar with this data set before you go into your exam, including some basic geographic knowledge!



Edexcel's data set concerns **weather data from a number of weather stations**. Let's explore what you might be expected to know. . .

https://qualifications.pearson.com/content/dam/pdf/A%20Level/Mathematics/2017/specification-and-sample-assesment/Pearson%20Edexcel%20GCE%20AS%20and%20AL%20Mathematics%20data%20set%20-%20Issue%201%20(1).xls





1 You should know the names and rough locations of the 5 UK weather stations, as well as the 3 international weather stations.

The data was recorded for:

- May-Oct 1987
- May-Oct 2015

2 You should be familiar with the variables involved and their respective units.

Total rainfall

(in mm)

tr/trace means less than 0.05mm

Mean Windspeed

kn/knot is "nautical mile per hour". $1kn = 1.15mph$

Windspeed also given on **Beaufort Scale**:

0 = Calm	$< 1kn$
1-3 = Light	1-10kn
4 = Moderate	11-16kn
5 = Fresh	17-21kn

Mean Visibility

How far (in metres) can be seen into the horizon during daylight hours.

Wind Direction

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	CAMBORNE		© Crown Copyright Met Office 2015										
2	NGR = 1627E 4067N												
3	Altitude = 87 metres												
4	Latitude = 50:22N Longitude = 05:33W												
5													
6	Date	Daily Mean Temperature (0900-0900) (°C)	Daily Total Rainfall (0900-0900) (mm)	Daily Total Sunshine (0000-2400) (hrs)	Daily Mean Windspeed (0000-2400) (kn)	Daily Mean Windspeed (0000-2400) (Beaufort conversion)	Daily Maximum Gust (0000-2400) (kn)	Daily Maximum Relative Humidity %	Daily Mean Total Cloud (oktas)	Daily Mean Visibility (Dm)	Daily Mean Pressure (hPa)	Daily Mean Wind Direction (o)	Cardinal Direction	Daily Max Gust Corresponding Direction (o)	Cardinal Direction
7	01/05/1987	10.7	3.1	n/a	n/a	n/a	n/a	100	7	2000	1018	360	N	20	NNE
8	02/05/1987	8.9	0.1	n/a	n/a	n/a	n/a	91	3	3200	1020	320	NW	330	NNW
9	03/05/1987	8.1	0	n/a	n/a	n/a	n/a	77	5	3600	1029	350	N	350	N
10	04/05/1987	8.2	0	n/a	n/a	n/a	n/a	83	5	4100	1036	350	N	350	N
11	05/05/1987	9.8	0	n/a	n/a	n/a	n/a	86	5	2700	1036	10	N	10	N
12	06/05/1987	9.3	0	n/a	n/a	n/a	n/a	100	1	1000	1033	330	N		
13	07/05/1987	10.9	0	n/a	n/a	n/a	n/a	100	3	600	1031	350			
14	08/05/1987	10.5	tr	n/a	n/a	n/a	n/a	89	1	2400	1025	110			
15	09/05/1987	10.9	0	n/a	n/a	n/a	n/a	95	3	900	1017	360			
16	10/05/1987	9.9	0	n/a	n/a	n/a	n/a	79	4	4100	1018	10			
17	11/05/1987	8.8	6	n/a	n/a	n/a	n/a	95	7	2500	1017	270	W	260	W
18	12/05/1987	10.2	tr	n/a	n/a	n/a	n/a	97	5	2400	1009	310	NW	310	NW
1			2.2	n/a	n/a	n/a	n/a	77	4	4600	1016	340	NNW	340	NNW
2			tr	5.9	16	Moderate	33	95	7	3100	1008	290	WNW	270	W
2			0	12.3	13	Moderate	27	77	4	4500	1012	10	N	10	N
2			tr	11.6	6	Light	16	92	4	3700	1015	290	WNW	290	WNW

Mean Pressure

In hectopascals (hPa)

Mean temperature

(in $^{\circ}C$)

Textbook claims this is **max** temp for UK, but it is **mean** temp for all locations.

Total sunshine

(nearest $\frac{1}{10}$ of an hour)

Maximum Gust

(in kn) is highest instantaneous wind speed.

Humidity

is the % of air saturation with water vapour. 100% is the maximum % water content air can contain.

Mean Cloud Cover

Oktas means the number of $\frac{1}{8}$ths of the sky covered.

3 You should have a vague idea of the range of values for each location.

UK Location (2015)	Temp Range	Wind Speed Range
Camborne	10-20	3-18
Heathrow	8-**29**	3-19
Hurn	6-24	2-19
Leeming	4-23	3-17
Leuchars	4-19	3-23

World Location (2015)	Temp Range	Wind Speed Range
Beijing	8-33	2-9
Jacksonville	15-31	1-12
Perth	8-25	4-14

Beijing temp range relatively large.
Min Jacksonville temp high.
Perth similar to UK.

Mean wind speed in UK across full period was roughly 9 nm. But 4 nm in Beijing (i.e. lower), 5 in Jacksonville (again lower), 8 in Perth (similar to UK).

From new A Level sample assessment materials:

"A meteorologist believes that there is a relationship between the daily mean windspeed, w kn, and the daily mean temperature, t °C. A random sample of 9 consecutive days is taken from past records from a town in the UK in July and the relevant data is given in the table below. . . .

Using the same 9 days, a location from the large data set gave $\bar{t} = \mathbf{27.2}$ and $\overline{w} = \mathbf{3.5}$.

(d) Using your knowledge of the large data set, suggest, giving your reason, the location that gave rise to these statistics."

(Note to teachers: I will not otherwise use SAM questions in these slides. I made one exception here!)

4 You should have a vague idea of the range of values for each variable for the data set as a whole.

Variable	Typical value(s)
Gust (UK only)	8 – 52 nm
Rainfall	0 – 60 mm in UK, but more extreme maximums elsewhere (e.g. 102mm in Perth)
Pressure	988 – 1038 hPa
Wind Speed on Beaufort scale	Max is 'fresh' (5). Most Light or Moderate.
Sunshine (UK only)	0 – 16 hrs
Cloud Cover	0 – 8 ocktas (i.e. full spread)

Hurn

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Daily maximum temperature and rainfall: 09:00–09:00. Sunshine, mean wind and maximum gust: 00:00–24:00.

June 1987	Max °C	Rain mm	Sun h	Wind kn	Beaufort	Gust kn
01/06	15.1	.6	4.5	7	Light	19
02/06	12.5	4.7	0	7	Light	22
03/06	13.8	tr	5.6	11	Moderate	25
04/06	15.5	5.3	7.8	7	Light	17
05/06	13.1	19	.5	10	Light	33
06/06	13.8	0	8.9	19	Fresh	46
07/06	13.2	tr	3.8	11	Moderate	27
08/06	12.9	1	1.7	9	Light	19
09/06	11.2	tr	5.4	6	Light	19
10/06	9.2	1.3	9.7	4	Light	n/a
11/06	12.6	0	12.5	6	Light	18
12/06	10.4	0	11.9	5	Light	n/a
13/06	9.6	0	8.6	5	Light	15
14/06	10.2	0	13.1	5	Light	18
15/06	9.2	3.7	7.1	4	Light	25
16/06	10.4	5.6	8.3	6	Light	25
17/06	12.8	.1	5.3	10	Light	27
18/06	13.0	7.4	3.2	9	Light	24
19/06	14.0	tr	.4	12	Moderate	33
20/06	12.6	0	7.7	6	Light	17

As previously noted, the **actual** data set has mean temperature for all locations. I changed to maximum temperature for this example for consistency with the textbook.

[Textbook]

(a) Describe the type of data represented by daily total rainfall.

Alison is investigating daily maximum gust. She wants to select a sample of size 5 from the first 20 days in Hurn in June 1987. She uses the first two digits of the date as a sampling frame and generates five random numbers between 1 and 20.

b) State the type of sample selected by Alison.

c) Explain why Alison's process might not generate a sample of size 5.

a a

b b

c c

Hurn

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Daily maximum temperature and rainfall: 09:00–09:00. Sunshine, mean wind and maximum gust: 00:00–24:00.

June 1987	Max °C	Rain mm	Sun h	Wind kn	Beaufort	Gust kn
01/06	15.1	.6	4.5	7	Light	19
02/06	12.5	4.7	0	7	Light	22
03/06	13.8	tr	5.6	11	Moderate	25
04/06	15.5	5.3	7.8	7	Light	17
05/06	13.1	19	.5	10	Light	33
06/06	13.8	0	8.9	19	Fresh	46
07/06	13.2	tr	3.8	11	Moderate	27
08/06	12.9	1	1.7	9	Light	19
09/06	11.2	tr	5.4	6	Light	19
10/06	9.2	1.3	9.7	4	Light	n/a
11/06	12.6	0	12.5	6	Light	18
12/06	10.4	0	11.9	5	Light	n/a
13/06	9.6	0	8.6	5	Light	15
14/06	10.2	0	13.1	5	Light	18
15/06	9.2	3.7	7.1	4	Light	25
16/06	10.4	5.6	8.3	6	Light	25
17/06	12.8	.1	5.3	10	Light	27
18/06	13.0	7.4	3.2	9	Light	24
19/06	14.0	tr	.4	12	Moderate	33
20/06	12.6	0	7.7	6	Light	17

[Textbook] Calculate:

a) The mean daily maximum temperature for the first five days of June in Hurn in 1987.

b) The median daily total rainfall for the week of 14th June to 20th June inclusive.

c) The median daily total rainfall for the same week in Perth was 19.00mm. Karl states that more southerly countries experience higher rainfall during June. State with a reason whether your answer to part (b) supports this statement.

a a

b b

c c

Questions 3 and 4 use the following extracts from the large data set.

Leeming · © Crown Copyright Met Office 2015

Date	Mean temp. (°C)	Total rain (mm)	Sunshine (h)	Mean wind (kn)
01/06/15	8.9	10	5.1	15
02/06/15	10.7	tr	8.9	17
03/06/15	12.0	0	10.0	8
04/06/15	11.7	0	12.8	7
05/06/15	15.0	0	8.9	9
06/06/15	11.6	tr	5.4	17
07/06/15	12.6	0	13.9	10
08/06/15	9.4	0	9.7	7
09/06/15	9.7	0	12.1	5
10/06/15	11.0	0	14.6	4

Heathrow · © Crown Copyright Met Office 2015

Date	Mean temp. (°C)	Total rain (mm)	Sunshine (h)	Mean wind (kn)
01/06/15	12.1	.6	4.1	15
02/06/15	15.4	tr	1.6	18
03/06/15	15.8	0	9.1	9
04/06/15	16.1	.8	14.4	6
05/06/15	19.6	tr	5.3	9
06/06/15	14.5	0	12.3	12
07/06/15	14.0	0	13.1	5
08/06/15	14.0	tr	6.4	7
09/06/15	11.4	0	2.5	10
10/06/15	14.3	0	7.2	10

3. Use the sunshine data for the first ten days of June 2015.

(a) Find the mean daily total sunshine for (i) Leeming and (ii) Heathrow.

(b) Find the range for (i) Leeming and (ii) Heathrow.

(c) Supraj says the further north you are, the fewer hours of sunshine. Do your answers support this? Give reasons.

3a(i)

3a(ii)

3b(i)

3b(ii)

3c

4. Calculate the mean daily total rainfall at Heathrow for the first ten days of June 2015. Explain how you treated 2, 5 and 8 June.

4

5. Dominic investigates how average monthly temperature changed during summer 2015 in Jacksonville. He averages temperatures for two days each month and compares the means.

(a) Give one reason two days a month might be (i) a good sample size, (ii) a poor sample size.

(b) He takes the first and last day of each month. Why might this be unrepresentative?

(c) Suggest a better way to choose days.

a(i)

a(ii)

b

c

7. Brian calculates mean cloud coverage in Leeming in September 1987 as 9.3 oktas. Explain why this is incorrect.

7

8. The large data set contains 184 consecutive days from 1987. Marie investigates daily mean windspeeds in Camborne.

(a) Describe how to take a systematic sample of 30 days. (3 marks)

(b) Why would this sample not necessarily give 30 data points for her investigation? (1 mark)

8a

8b